Lab 1 - Multivariable Scatterplots

Goals and Learning Objectives:

By completing this Lab you will learn how to:

- Utilize Rtutor.ai to create multivariable scatterplots
- Identify patterns and trends in multivariable scatterplots
- Interpret estimates of correlation and association between two attributes (i.e., strength and direction of a correlation)
- Evaluate the real world implications of differences in trends between two quantitative attributes across different groups

Lab Introduction:

In this Lab, you will be guided through using RTutor.ai to create and interpret a multivariable scatterplot based on a fictional real world context. You will then be given the choice of four contexts to use to complete a second analysis. Your assignment is to choose one of those four contexts and answer the question posed.

To successfully complete this lab, you must submit a pdf that includes your answers to the following components of this lab:

1. Tutorial Part 1: In no more than one page, include your drawing and answers to the questions provided.

2. Tutorial Parts 2 and 3: In no more than one page, include your final prompt, and your answers to the Part 3 questions. On a second page, include a screenshot of your final prompt and the output RTutor.ai generates based on your prompt.

3. Tutorial Part 4: In 250-300 words, answer the question provided. Sufficiently justify and defend your decision utilizing the claims-evidence-reasoning model for argumentation.

4. Assignment: In no more than one page, include a 50-100 word description of the topic you chose and why, and a 250-300 word answer to the question provided. On a second page, include your final prompt, and a screenshot of your final prompt and the output RTutor.ai generates based on your prompt.

Your final submission should be a pdf file of exactly 5 pages in length. Utilize Times New Roman font, single spaced, size 12. Add a title in bold to each page. A template will be provided for you to utilize to help you in preparing your submission.

Tutorial Instructions:

A couple is moving to Chicago and is seeking the best-rated public high school for their child. Your task is to research and help them determine which school would be the best choice.

You have identified the following three top-rated public high schools in Chicago:

1. Walter Payton College Preparatory High School
2. Northside College Preparatory High School
3. Whitney M. Young Magnet High School

You have data on ACT scores and the number of hours spent preparing for the test for 18 different students from these schools. Based on this information, your task is to create a graph that will help you provide a recommendation on which school the couple should choose for their child. Note: ACT scores are scaled from a minimum of 0 to a maximum of 36. The median score is approximately 20, with the top 10% of students scoring above 30.

Tutorial Part 1: Plan a visualization

Take a few minutes to think about the kind of graph you would create to help you make a recommendation to the couple. Sketch this graph as best you can on a piece of paper – you don't need to use the real data points themselves, but consider what you think would be a realistic graph, and consider the graph's features.

How will this graph help you provide a recommendation on which school might be the best choice based on the data?

Describe each of the features of the graph that you sketched. Explain why each feature you included is helpful in providing a recommendation.

Tutorial Part 2: Create a multivariate scatter plot in Rtutor.ai

We would like you to visualize the relationship between the amount of time students spend preparing for the ACT and their final ACT scores across the three different high schools.

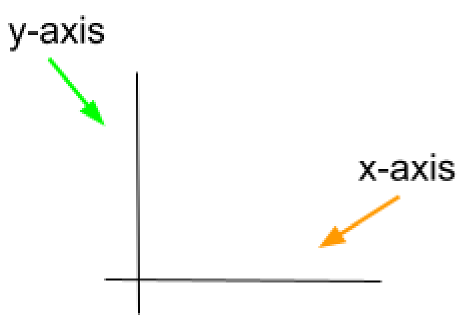
First, load the data set into RTutor.ai:

1. Download the data file here
    a. Each observation in the data set corresponds to one student.
    b. There are three attributes in the data set:
        i. school = the name of the high school the student attends
        ii. act = the student's ACT score
        iii. hrs = the number of hours that the student spent preparing for the ACT
2. Load the csv file into RTutor.ai.

Step 1: Telling Rtutor.AI to make a scatterplot

The first step in making a scatterplot in RtutorAI is simply to tell RtutorAI that is what we want.

At the same time, we also have to specify which variables go on which axis. We typically refer to the axes as the "Y-axis" and the "X-axis", as seen in the figure below:



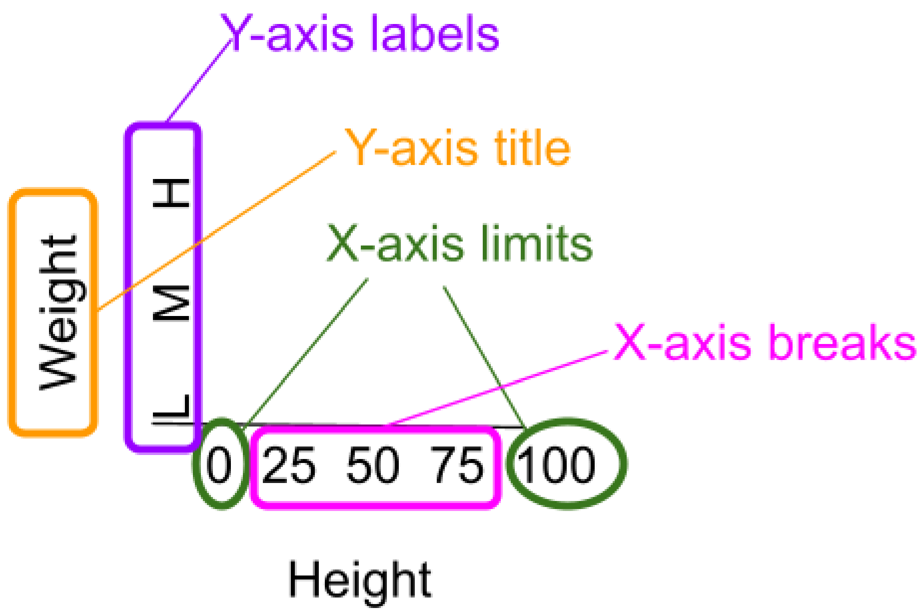
Tell RtutorAI which variable you want to put on each axis by saying something like this:

Make a scatterplot of the data.
Put the 'height' variable on the x-axis, and the 'weight' variable on the y-axis

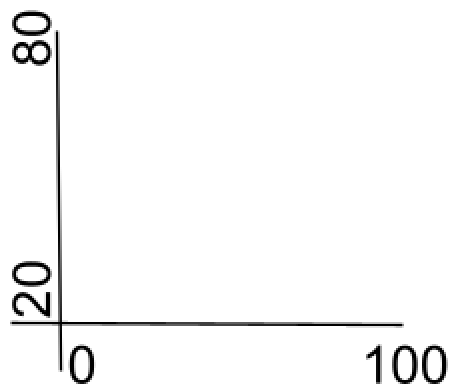
Now you try! Write a prompt to create a scatterplot of the Chicago Public School Data. What are the variables you want to put on each axis? Explain why you chose to put each variable on its respective axis.

Step 2: Changing Axis Limits, Breaks, Labels, and Titles

RtutorAI allows us to customize the graph however we want. There are four main ways we customize axes: changing the limits, breaks, labels, and titles. For example, the scatterplot below has x-axis limits from 0 to 100, x-axis breaks at 0, 25, 50, 75, and 100, an x-axis title that says “Height”, y-axis limits from 20 to 80, y-axis breaks at 20, 50, and 80, y-axis labels of “L”, “M”, and “H” for low, medium, and high, and a y-axis title that says “Weight”.



Axis Limits Axis Limits specify the starting and ending points of the axis. For example, in the scatterplot below, the x-axis limits are 0 and 100, while the y-axis limits are 20 and 80.



Question: What considerations would you take into account when choosing axis limits?

Tell RtutorAI what you want the axis limits to be by saying something like this:

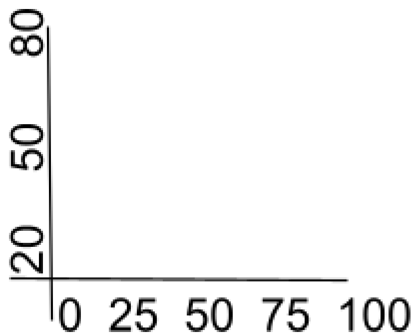
> Make a scatterplot of the data.
> Put the 'height' variable on the x-axis, and the 'weight' variable on the y-axis
> Make the x-axis limits 0 to 100, and the y-axis limits 20 to 80.

Now you try! Add to your previous prompt. Specify the axis limits for RtutorAI.

Hint: If the axis limits are too small, you can change the font sizes of the axis. You can set the font size for the entire graph all at once, or separately for each part of the graph. You can also set any font size you'd like. Add the prompt "make all font sizes 20" to increase the size of all text in the plot.

Reflection: How did RtutorAI select axis limits by default? Explain how you chose to change them, or explain why you chose to stick with the default limits.

Axis Breaks Axis Breaks are all the points in-between the axis limits. For example, in the scatterplot below, the x-axis breaks are 0, 25, 50, 75, and 100, while the y-axis breaks are 20, 50, and 80.



Question: What considerations would you take into account when choosing axis breaks?

Tell RtutorAI what you want the axis limits to be by saying something like this:

> Make a scatterplot of the data.
> Put the ‘height’ variable on the x-axis, and the ‘weight’ variable on the y-axis
> Make the x-axis limits 0 to 100, and the y-axis limits 20 to 80.
> Set x-axis breaks at 0, 25, 50, 75, and 100, and set y-axis breaks at 20, 50, and 80.

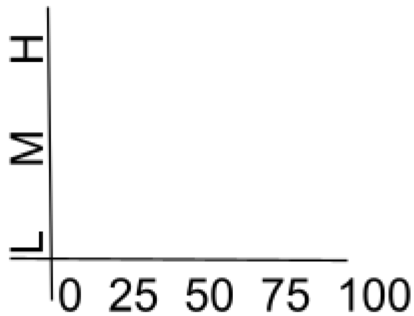
Now you try! Add to your previous prompt. Specify the axis breaks for RtutorAI.

Hint: If your breaks are evenly spaced, you can make your prompt more efficient by specifying the distance between each break:

> Make a scatterplot of the data.
> Put the ‘height’ variable on the x-axis, and the ‘weight’ variable on the y-axis
> Make the x-axis limits 0 to 100, and the y-axis limits 20 to 80.
> Set x-axis breaks every 25, and set y-axis breaks every 30.

Reflection: How did RtutorAI select axis breaks by default? Explain how you chose to change them, or explain why you chose to stick with the default breaks.

Axis Labels Axis Labels are labels for all the axis breaks. For example, in the scatterplot below, the y-axis labels are L, M, and H, for low, medium, and high.



Question: What considerations would you take into account when choosing axis labels?

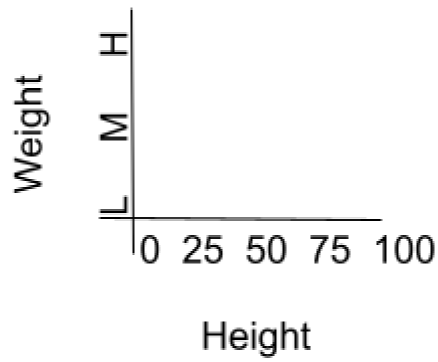
Tell RtutorAI what you want the axis labels to be by saying something like this:

> Make a scatterplot of the data.
> Put the ‘height’ variable on the x-axis, and the ‘weight’ variable on the y-axis
> Make the x-axis limits 0 to 100, and the y-axis limits 20 to 80.
> Set x-axis breaks at 0, 25, 50, 75, and 100, and set y-axis breaks at 20, 50, and 80.
> Make the y-axis labels “Low”, “Medium”, and “High”

Now you try! Add to your previous prompt. Specify the axis labels for RtutorAI.

Reflection: How did RtutorAI select axis labels by default? Explain how you chose to change them, or explain why you chose to stick with the default labels.

Axis Titles Axis Titles are titles for each axis. For example, in the scatterplot below, the x-axis title is “Height”, and the y-axis title is “Weight”.



Question: What considerations would you take into account when choosing axis titles?

Tell RtutorAI what you want the axis titles to be by saying something like this:

> Make a scatterplot of the data.
> Put the ‘height’ variable on the x-axis, and the ‘weight’ variable on the y-axis
> Make the x-axis limits 0 to 100, and the y-axis limits 20 to 80.
> Set x-axis breaks at 0, 25, 50, 75, and 100, and set y-axis breaks at 20, 50, and 80.
> Make the y-axis labels “Low”, “Medium”, and “High”
> Make the x-axis title “Height” and the y-axis title “Weight”.

Now you try! Add to your previous prompt. Specify the axis titles for RtutorAI.

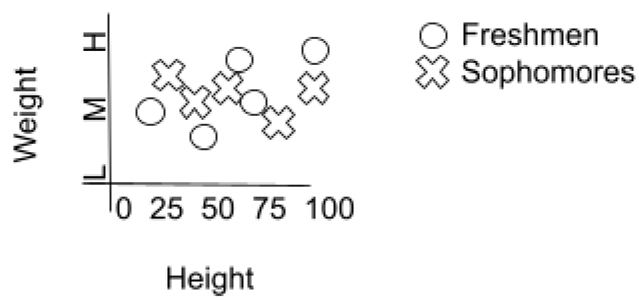
Hint: It is also helpful to add a main title to the plot. Add a main title to the plot by adding a line to your prompt like this: Make the main plot title ‘My First Scatterplot with Rtutor.AI”.

Reflection: How did RtutorAI select axis titles by default? Explain how you chose to change them, or explain why you chose to stick with the default titles.

Step 3: Adding distinct colors and shapes for each group

RtutorAI allows us to customize the graph however we want. When we have observations from multiple groups, it is helpful to distinguish them in some way. The two main ways we distinguish observations from different groups in a scatterplot is by changing the shape of each point (i.e., giving observations from different groups different shapes), or by changing the color of each point (i.e., giving observations from different groups different colors).

Point Shapes Point shapes change the shape of each point (i.e., each observation) on the scatterplot based on the observation's membership in some group. For example, in the scatterplot below, there are two groups of people, Freshmen - denoted by the shape "X", and Sophomores - denoted by the shape "O". Note that there is also a legend that appears with the graph which helps specify which shape is associated with which group.



Question: What considerations would you take into account when choosing point shapes?

Tell RtutorAI you want the point shapes to be different for each group by saying something like this:

> Make a scatterplot of the data.
> Put the 'height' variable on the x-axis, and the 'weight' variable on the y-axis.
> Change the shape of each point based on the 'year-in-school' variable.

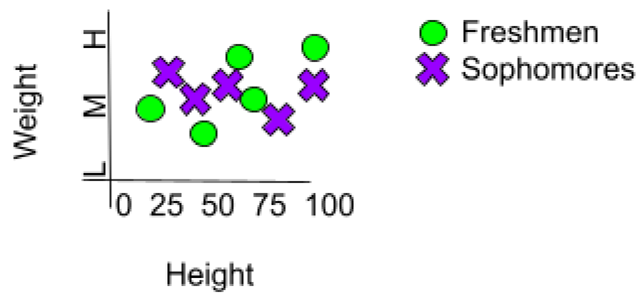
Now you try! Add to your previous prompt. Specify point shapes for RtutorAI.

Hint: The prompt above does not list specific shapes. You can tell RtutorAI that you want specific shapes for specific groups, although it is not required.

- *Don't forgot to add axis limits, breaks, labels, and titles, as well as a main plot title!*
- *You can move the legend to either the 'top' of the plot, 'left', 'right', or 'bottom'. Use a prompt like this: 'Place the legend at the bottom of the plot'.*

Reflection: How did RtutorAI select point shapes by default? Explain how you chose to change them, or explain why you chose to stick with the default shapes.

Point Colors Point colors change the color of each point (i.e., each observation) on the scatterplot based on the observation's membership in some group. For example, in the scatterplot below, there are two groups of people, Freshmen - denoted by color green, and Sophomores - denoted by the color red.



Question: What considerations would you take into account when choosing point colors?

Tell RtutorAI you want the point colors to be different for each group by saying something like this:

> Make a scatterplot of the data.
> Put the 'height' variable on the x-axis, and the 'weight' variable on the y-axis.
> Change the shape of each point based on the 'year-in-school' variable.
> Change the color of each point based on the 'year-in-school' variable.

Now you try! Add to your previous prompt. Specify point colors for RtutorAI.

Hint: The prompt above does not list specific colors. You can tell RtutorAI that you want specific colors for specific groups, although it is not required.

Hint: Don't forget to add axis limits, breaks, labels, and titles, as well as a main plot title!

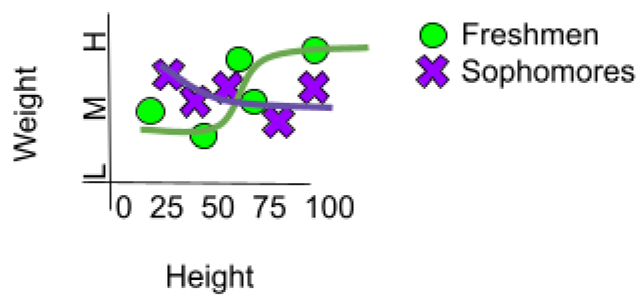
Hint: A good practice when using colors is to tell RtutorAI to "Use a color-blind friendly color scheme".

Hint: In this example, we used the same variable for both shape and color, but this is not required. You can use different group variables for shapes and colors if you want.

Reflection: How did RtutorAI select point colors by default? Explain how you chose to change them, or explain why you chose to stick with the default colors.

Step 4: Adding smoothed trend lines

When using a scatter plot to interpret trends and relationships between variables, it is very helpful to add a smoothed trend line. This is especially true when we want to compare trends between groups. For example, in the scatterplot below, there are two groups of people, Freshmen and Sophomores, and each group has their own trend line for the relationship between height and weight within the group.



Question: What considerations would you take into account when adding a smoothed trend line?

Tell RtutorAI you want to add a smoothed trend line for each group by saying something like this:

> Make a scatterplot of the data.
> Put the ‘height’ variable on the x-axis, and the ‘weight’ variable on the y-axis.
> Change the color of each point based on the ‘year-in-school’ variable.
> Add a smoothed trend line for each group.

Now you try! Add to your previous prompt. Tell RtutorAI to add a smoothed trend line.

Hint: Smoothed trend lines are supposed to be curvy. If RtutorAI creates a straight line, add the following text to your prompt about the smoothed trend line: “use the loess method for the trend line”.

Hint: You can incorporate uncertainty and ish-ness in the trend line by adding what is known as an error region around the line. Add the prompt: “include the error region around the trend line”.

Reflection: How did RtutorAI add the smoothed trend line by default? Explain how you chose to change the default options, or explain why you chose to stick with the default options.

Tutorial Part 3: Interpret the features of the graph

Compare the graph you created in Rtutor.ai with your initial sketch.

How do the graphs differ?

Which features of the graph did you draw that are not a part of the Rtutor.AI graph?

Which features of the graph help you identify and interpret patterns in the data and differences between the schools?

Now you try! Add to your previous prompt. Try adding additional features to the graph.

Tutorial Part 4: Draw a conclusion

Which school do you recommend the couple send their child to? Explain your reasoning based on the data visualized in RTutor.ai. Justify your choice by referring to specific statistical features of the graph and your interpretation of what they mean in the real world.

Assignment: Conduct another analysis

In this part of the activity, you will complete an analysis of a topic and data set of your choice. The choices are:

(1) Psych related question
(2) Medical related question
(3) Business related question
(4) Sports related question

Choose your topic, download the appropriate data set, create a scatter plot, and use the scatter plot to help you answer the topic question.

Assignment Instructions (Medical)

Medical researchers want to understand how exercise affects body mass index (BMI) for different age groups. BMI is a number that helps measure whether someone is underweight, at a healthy weight, or overweight based on their height and weight. A BMI between 18.5 and 24.9 is generally considered a healthy range.

You have data on a group of patients that includes:

- Weekly minutes of moderate exercise (how much time they spend exercising per week)
- BMI
- Age group (one of four groups: 18-34, 35-39, 40-44, and 45+)

Your task is to create a graph investigating whether weekly exercise is related to BMI, as well as whether this relationship changes based on age group.

Download the data file here.

Assignment Instructions (Business)

A video game company wants to understand how critic scores affect total game sales for different types of games. Your task is to analyze the relationship between critic scores and sales to help the company make better decisions.

You have data on the average critic score (rated out of 10) and total sales (in millions of copies) for video games from three different genres: fighting, puzzle, and platform.

Your task is to create a graph investigating whether critic scores are related to sales, as well as whether this relationship changes based on genre.

Download the data file here.

Assignment Instructions (Psychology)

A group of psychologists is interested in understanding how self-efficacy impacts depression for clients with different personalities. They used scales to measure their clients' self-efficacy and depression, as well as a personality test. They provided you the following data:

- Self-efficacy scale ranges from 1 to 4 with a higher number meaning more self-efficacy (mean of 2.50)
- Depression scale ranged from 0 to 27 with a higher number meaning more depression depression (with a mean of 12)
- Personality categorized as either introvert or extrovert

Your task is to create a graph investigating whether self-efficacy is related to depression, as well as whether this relationship changes based on personality.

Download the data file here

Assignment Instruction (Sports)

A sports investment group is interested in buying a soccer league. However, they know that athletes' performance is affected by their age. Your task is to help them decide which soccer league is best to buy accounting for players' aging. You have the following data:

- Soccer leagues: Premier (England), La Liga (Spain), and Ligue1 (France)
- Goals scored per season ranging from 30 to 83 (with a mean of 51)
- Player ages ranging from 18 to 36 (with a mean of 27)

Your task is to create a graph investigating whether players' age is related to goals scored, as well as whether this relationship changes based on league.

Download the data file here